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MIL-L-49427(CR)
6 November 1989

MILITARY SPECIFICATION

LENS ASSEMBLY, EYEPIECE FOR AVIATOR'S NIGHT VISION IMAGING SYSTEM, AN/AVS-6(V)1, AN//AVS-6(V)2

This specification is approved for use by USACECOM, Department of Army, and is available for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Lens Assembly, Eyepiece, referred to herein as the eyepiece, which is part of the Aviator's Night Vision Imaging System, AN/AVS-6(V)1, AN/AVS-6(V)2, referred to herein as ANVIS (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.g).

SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials and Processes Used in Electronic Equipment
MIL-O-13830	-	Optical Components for Fire Control Systems

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.1 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

5002760	-	Image Intensifier Assembly, 18mm, MX-10160/AVS-6
5005836	-	Lens Assembly, Eyepiece
5005837	-	Eyepiece Cell Assembly

(Copies of specifications, standards, drawings and publications, required by the Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, supersedes applicable laws and regulations unless a specified exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. The eyepiece shall be constructed in accordance with 5005836 and as specified herein.

3.2 Qualification. Eyepiece lens furnished under this specification shall be products which are authorized by the Government as Qualified Products (see 3.9.5).

3.2.1 Initial production testing. When required by the contract the supplier shall furnish and test eyepieces as specified herein (see 6.2.b).

3.3 Parts and materials. Parts and materials shall be as specified herein and as shown on 5005836. Materials not specified shall be selected by the contractor and shall meet the requirements of MIL-P-11268.

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3.4 Performance characteristics. The eyepiece shall be designed and fabricated to operate with the output of the Image Intensifier Assembly, 18mm, MX-10160/AVS-6 in accordance with 5002760. The output fiber optic twist will have a 40.0mm, ± 0.1mm concave radius of curvature. Unless otherwise specified, the following parameters shall be met at room temperature (see 3.9.3).

3.4.1 Equivalent focal length. The equivalent focal length (EFL) of the eyepiece shall be 27.03, ± 0.50mm.

3.4.2 Flange focal distance. The flange focal distance of the eyepiece shall be 10.0 (+0.26 -0.40)mm, where the measurement is the distance between the rearmost projection of the eyepiece and the on-axis infinity focal plane.

3.4.3 Exit pupil vs. eye relief. The minimum flange eye relief shall be 12mm. Eye relief from the vertex of the eye to the vertex of the nearest eyepiece optical element surface along the optical axis shall be 15, (+0.0 -0.2)mm. Exit pupil is defined as the diameter of a circle at the eye relief distance within which the eye is presented good optical performance. The diameter of the exit pupil for an on-axis image shall be 14mm. Off-axis exit pupil for field images may approach a minimum of 7mm at a 15mm eye relief. This full field exit pupil shall be larger than 7mm at any shorter eye relief.

3.4.4 Linear distortion. The linear distortion of the eyepiece at the edge of the 18mm format shall be 8, ± 0.5 percent pincushion. This distortion accounts for the 40mm concave radius of curvature of the image.

3.4.5 Transmission. The transmission of the eyepiece shall be not less than 75 percent over the entire exit pupil as specified by 3.4.3 for spectral distribution defined by Figure 1.

3.4.6 Field of view (FOV). The field of view of the eyepiece shall be 40, ± 1 degrees for the 18mm format with a 40mm concave radius image.

3.4.7 Field curvature. The field curvature shall deviate no more than +1/4, -1/3 diopter over an 18mm format with a 40mm concave radius of curvature. The focal plane shall be defined as the average between the radial and tangential focal surfaces.

3.4.8 Concentricity. The image displacement of the mechanical and optical axis of the eyepiece shall not exceed 0.1mm.

3.4.9 Modulation transfer function (MTF). For the spectral distribution defined in Figure 1 the eyepiece shall have an MTF as follows.

3.4.9.1 Centered pupil MTF. The modulation transfer function of the eyepiece shall not be less than that specified in Figure 2 for a 5mm pupil centered at a 15mm eye relief.

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3.4.9.2 Decentered pupil MTF. The modulation transfer function of the eyepiece shall not be less than that specified in Figure 3 for a 5mm exit pupil at a 15mm eye relief and decentered as shown.

3.4.10 Weight. The weight of the eyepiece lens assembly (5005836) shall not exceed 64 grams. The weight of the eyepiece cell assembly (5005837) (see 3.9.4) shall not exceed 48 grams.

3.5 Environmental characteristics. When the eyepiece is incorporated into the ANVIS, it shall not cause the ANVIS to fail environmental requirements. The change in eyepiece optical performance with temperature shall not cause the ANVIS to fail specified performance requirements.

3.5.1 Vibration. The eyepiece shall not be damaged (see 3.9.1) after vibration in the frequency range of 5 to 500 hertz at an amplitude level of 0.2-inch double-amplitude or 2-g acceleration, whichever is smaller.

3.5.2 Temperature-altitude. The eyepiece shall not be damaged (see 3.9.1) during storage at temperatures ranging from -35°C to +71°C, altitudes ranging from sea level to 50,000 feet above sea level at a temperature of -35°C.

3.5.3 Temperature shock. The eyepiece shall not be damaged by sudden temperature changes over the temperature range of +71°C to +23°C and +23°C to -35°C (see 3.9.1).

3.5.4 Humidity. The coated eyepiece shall not be damaged nor suffer any degradation in performance when exposed to high humidity (95%) with varying temperature for prolonged exposure.

3.6 Treatment and painting. The finishes on the eyepiece shall be in accordance with 5005836. Any metal parts shall have finishes which are in compliance with MIL-STD-171.

3.7 Marking. The eyepiece shall be marked in accordance with 5005836.

3.8 Workmanship. All eyepieces shall be free from cracks, splits, cold flow, shrinkage, inclusions, porosity, or any similar characteristics. Threads shall be full and undamaged for the entire length or depth as required on the applicable drawing. Eyepieces shall be free from burrs and shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material. The cleaning methods used shall not damage any of the eyepieces, nor shall the parts be contaminated by the cleaning agent. Quality requirements defined as scratches, digs, edge chips, bubbles, coating defects and cementing defects shall be in accordance with MIL-O-13830 except that inspection of optical cleanliness shall be with the unaided eye (without 3X magnification). All moving parts and adjustments shall be examined to insure that they move freely throughout their entire range without sticking, binding, or creeping.

3.9 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

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3.9.1 Damage. Damage is defined as a failure of any finish; condensation, fogging or moisture on any internal surface; dents or breakage of any hardware, connection, or component coating blistering, peeling or milky conditions.

3.9.2 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

3.9.3 Room temperature. Room temperature is defined as 23°C, +10°C, -2°C.

3.9.4 Eyepiece lens cell. The eyepiece lens cell is defined as the cylindrical part which is in direct contact with the lens elements, holds the lens elements in their fixed relative position, and has no external sleeves, flanges, or other surfaces or fittings permanently attached to it in any manner.

3.9.5 Qualified product. A product which has successfully met all of the requirements of an IPT or first article witnessed by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the Contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the Contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections and tests. Inspections and tests shall be classified as follows:

a. Initial production testing (4.3)
b. Quality conformance inspections and tests (4.4)

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4.3 Initial production testing (IPT), inspections and tests. Unless otherwise specified (see 6.2.b) fifteen eyepieces manufactured with production tooling shall be provided for IPT. Each eyepiece shall be inspected in accordance with Table I and shall be subjected to Table II, subgroup 1 tests. The tests in subgroup 1 may be performed in any order except for MTF. For those eyepieces subjected to subgroups 1, MTF shall be performed last. Three eyepieces which pass subgroup 1 may be subjected to Table II, subgroup 2 tests. Failure of any inspection or test of Table I or Table II may be cause for rejection of IPT. Molded parts shall be chosen so as to have equal representation from all molds. Eight (8) samples which have passed subgroup 1 tests shall be subjected to subgroup 3, Humidity.

TABLE I. Inspections.

Requirement	Method
Workmanship (see 3.8)	Visual
Parts missing/damaged	Visual
*Packaging, missing or not as specified	Visual
Marking	Visual

*Final examination only

TABLE II. Initial production tests.

Requirement	Requirement paragraph	Test paragraph
Subgroup 1		
Equivalent focal length	3.4.1	4.5.1
Flange focal distance	3.4.2	4.5.2
Exit pupil vs. eye relief	3.4.3	4.5.3
Linear distortion	3.4.4	4.5.4
Transmission	3.4.5	4.5.5
Field of view	3.4.6	4.5.6
Field curvature	3.4.7	4.5.7
Concentricity	3.4.8	4.5.8
MTF	3.4.9	4.5.9
Weight	3.4.10	4.5.10
Subgroup 2		
Vibration	3.5.1	4.6.1
Temperature-altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3
Subgroup 3		
Humidity	3.5.4	4.6.4

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4.4 Quality conformance inspections and tests.

4.4.1 Inspections. All eyepieces shall be inspected for the attributes listed in Table I. Failure to meet requirements specified shall be cause for rejection of the eyepiece inspected.

4.4.2 Tests. For purposes of group B and group C testing, a lot shall be as defined by the contractor.

4.4.2.1 Group A tests. All eyepieces shall be tested for the requirements listed in Table III. Failure of an eyepiece to meet requirements specified shall be cause for rejection of that eyepiece.

TABLE III. Group A tests.

Requirement	Requirement paragraph	Test paragraph
Flange focal distance	3.4.2	4.5.2
MTF, centered pupil	3.4.9.1	4.5.9.1

4.4.2.2 Group B tests. Eyepieces which have passed the inspections and tests of Tables I and II shall be tested in accordance with Table IV. Sampling for inspection shall be as described in the contract or purchase order (see 6.2.e).

TABLE IV. Group B tests.

Requirement	Requirement paragraph	Test paragraph
Subgroup 1		
Concentricity	3.4.8	4.5.8
Subgroup 2		
Transmission	3.4.5	4.5.5
Weight	3.4.10	4.5.10

4.4.2.3 Group C tests. Lots which have passed the group B tests shall be tested in accordance with Table V. Sampling for inspection shall be as described in the contract or purchase order (see 6.2.e).

4.4.2.3.1 Group C failure actions. Actions required in the event of a Group C failure shall be as specified in the contract (see 6.2.c).

4.4.2.3.2 Group C samples disposition. Group C samples shall be refurbished, pass Group A inspection and shipped as part of the contract.

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TABLE V. Group C tests.

Requirements	Requirement paragraph	Test paragraph
Subgroup 1		
Equivalent focal length	3.4.1	4.5.1
Exit pupil vs. eye relief	3.4.3	4.5.3
Linear distortion	3.4.4	4.5.4
Field of view	3.4.6	4.5.6
Field curvature	3.4.7	4.5.7
MTF, decentered pupil	3.4.9.2	4.4.9.2
Subgroup 2		
Vibration	3.5.1	4.6.1
Temperature-altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3
Subgroup 3		
Humidity	3.5.4	4.6.4

4.5 Test methods. Unless otherwise specified, all tests shall be performed at room temperature. Temperature chambers shall maintain specified temperatures within 2.0°C. The tolerances specified herein are absolute, with no allowance for test equipment inaccuracy. The tolerances used by the contractor shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment. All optical tests, with the exception of exit pupil/eye relief, are performed with the eyepiece run backwards (i.e., collimated light enters the eye lens). Unless otherwise specified, all tests are performed with a 5 ± .2mm stop placed on-axis at a 15mm, +0, -0.2 eye relief.

4.5.1 Effective focal length. The eyepiece lens assembly with a simulated eye pupil of five (5) mm diameter at the eyerelief distance shall be placed on a goniometric table that includes a measuring microscope. The light source is a collimated beam, with a pinhole or slit target, of sufficient bundle size as to fill the lens under test at all possible field angles. The size of the pinhole/slit width as geometrically projected through the lens under test shall not be greater than 0.005mm. The microscope measuring ways are previously aligned to the point source or slit source target. The effective focal length (EFL) for the eyepiece lens assembly is determined by measuring the image translation produced by rotating the goniometer though an angle of ± one (1) degree. The effective focal length is the quotient of the difference in the measuring microscope readings divided by two over the tangent of one degree. Failure to meet the requirements of 3.4.1 shall constitute failure of this test.

4.5.2 Flange focal distance. The flange focal distance of the eyepiece shall be measured as part of the center pupil MTF test, 4.5.9.1. Failure to meet requirements of 3.4.2 shall constitute failure of this test.

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4.5.3 Exit pupil vs. eye relief. Using a diopterscope set for 0 diopter, center an 18.0, (+0.1mm -0.0)mm 40mm radius concave back-illuminated target at the infinity focal plane, (this simulates the image tube). Position an opaque mask at 15.0mm, (+0.0mm -0.2)mm eye relief distance. The mask shall have two pinholes (1mm diameter) 7mm apart (center to center) and centered about the optical axis. View through each pinhole to determine if the entire 18mm target is visible through each pinhole. The opaque mask may not be moved between viewings. Failure to observe the entire 18mm target through both pinholes shall constitute failure of this test. Failure to meet the requirements of 3.4.3 shall constitute failure of this test.

4.5.4 Linear distortion. The eyepiece lens assembly with a simulated eye pupil of five (5) mm diameter at the eyerelief distance shall be placed on a goniometric table that includes a measuring microscope. The light source is a collimated beam, with a pinhole or slit target, of sufficient bundle size as to fill the lens under test at all possible field angles. The size of the pinhole/slit width as geometrically projected through the lens under test shall not be greater than 0.005mm. The microscope measuring ways are previously aligned to the point source or slit source target. The linear distortion for the eyepiece lens assembly is determined by measuring the angular rotation of the goniometer required for an image height of 9mm on a 40mm radius of curvature. The percent distortion is 100 *(((afh/tan (afa))-EFL)/EFL). The variables EFL is effective focal length, afa is average semi field angle (e.g. 20 degrees), afh is average semi field height (e.g. 9mm). Failure to meet requirements of 3.4.4 shall constitute failure of this test.

4.5.5 Transmission. Transmission of the eyepiece shall be measured in accordance with MIL-STD-150, Method 5 or 6. A 10mm stop shall be placed on axis in front of the lens under test. The light source and detector combination shall yield a spectral response curve complying with Figure 1. Failure to meet requirements of 3.4.5 shall constitute failure of this test.

4.5.6 Field of view. The field of view of the eyepiece shall be calculated using the data obtained for a 9mm semifield height in the linear distortion test 4.5.4. Failure to meet requirements of 3.4.6 shall constitute failure of this test.

4.5.7 Field curvature. Field curvature of the eyepiece shall be measured in accordance with MIL-STD-150, Method 16. A measuring microscope shall be used in place of a photographic plate. In order to use this procedure, the eyepiece shall be run backwards (i.e., collimated light enters the eye lens) with a 5.0 mm, ± 0.2 mm exit pupil at a 15.0, (+0.0, -0.2)mm eye relief distance between the collimator and the eyepiece. Separate measurements shall be taken for radial and tangential targets. The plane at best focus shall be defined as the average between the two planes. Failure to meet requirements of 3.4.7 shall constitute failure of this test.

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4.5.8 Concentricity. The mechanical axis of the eyepiece lens cell is controlled by the threads on the outside of eyepiece cell assembly (5005837) and by the O-ring. The optical axis shall then be located for an image lying on the mechanical axis at infinity, and the variation in the optical axis from the mechanical axis shall be measured. The eyepiece cell assembly shall be mounted on a fixture that engages the treads and the O-ring on the assembly. A bearing surface which rotates the assembly about its mechanical axis shall be concentric to within 0.025mm. Failure to meet the requirements of 3.4.10 shall constitute failure of this test.

4.5.9 Modulation transfer function (MTF). The eyepiece shall be tested on an MTF apparatus that yields sine-wave MTF response. The MTF apparatus shall have a light source and detector, the combination of which yields a spectral response curve complying with Figure 1. (If a microscope objective is used to relay the image of the eyepiece, the numerical aperture (NA) of the microscope objective shall be not less than 0.45.) The MTF apparatus shall allow the eyepiece to be tested either continuously or at a minimum of 4 distinct, evenly distributed points from 0 to 40 lp/mm. The area analyzed shall be at least 0.5 mm in diameter. The calibration of the MTF apparatus shall be performed before the test data is valid. The calibration procedure to be used is to measure the MTF of a known size target (e.g. slit or pinhole). All references to size are at second surface of the simulated photocathode. The MTF of a pinhole is known mathematically as a first order Bessel Function. MTF = $(2J_1(x))/x$ where x = 3.14156* (lp/mm)*(pinhole diameter in mm). The MTF of a pinhole goes to zero at a spatial frequency 1.22 times its reciprocal diameter. The MTF of a slit is known mathematically as (sine (x))/ (x), where x = 3.14156* (lp/mm)*(slit width). The units for x is in radians. The MTF of a slit goes to zero at a spatial frequency at its reciprocal width. The procedure shall be to measure the MTF of the lens and then measure the MTF of the lens with a very large pinhole/slit. In the case of microprocessor based MTF apparatus which correct for the pinhole/slit size, the size of the pinhole/slit entered into the computer software, when measuring the lens with large pinhole/slit should not be greater than 1/40 of the larger pinhole/slit. The geometrical size of the aerial image shall be 0.030mm +/-0.001. The geometrical size of the slit shall be 0.025 +/- 0.001. The MTF of the large pinhole/slit can be determined by taking the ratio of the MTF curves of the lens with large pinhole/slit, divided by the MTF of the lens under test. Differences in Modulation of greater than three percent will invalidate the MTF results. See Figure 3 for MTF as a function of spatial frequency and pinhole diameter/slit width.

4.5.9.1 Centered pupil MTF. Place a 5.0, ± 0.2 mm stop at a 15.0, (+0.0 -0.2) mm eye relief distance centered with respect to the eye lens aperture (i.e., on the optical axis) and measure the on-axis MTF. Rotate the lens and analyzer to the 2/3 field height position and measure radial and tangential MTF. Off-axis measurements need not be in the same plane chosen for on-axis measurements. Failure to meet requirements in 3.4.9.1 shall constitute failure of this test.

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4.5.9.2 Decentered pupil MTF. Place a 5.0, ± 0.2mm stop at the 15.0, (+0.0 -0.2)mm eye relief distance centered with respect to the eye lens aperture (i.e., on the optical axis). Measure the on-axis MTF of the eyepiece versus exit-pupil decenteration from the optical axis. Measurements shall be made both parallel to and perpendicular to the long dimension of a split image. Rotate the lens and analyzer to the 2/3 field height position and measure off-axis MTF versus decentered pupils for both radial and tangential targets. Each MTF measurement may be refocused for maximum response. Failure to meet requirements of 3.4.9.2 shall constitute failure of this test.

4.5.10 Weight. The eyepiece shall be weighed on a balance or scale accurate to within ± 0.1 gram. Failure to meet requirements of 3.4.10 shall constitute failure of this test.

4.6 Environmental tests. Each eyepiece shall demonstrate its ability to perform properly after being subjected to the Group C, subgroup 2 or subgroup 3 tests. For the purpose of post-environmental testing, "perform properly" shall be defined as the ability to successfully complete, in order, modulation transfer function (4.5.9), transmission (4.5.5), and flange focal distance (4.5.2) tests.

4.6.1 Vibration. The eyepiece shall be mounted to a test fixture and subjected to harmonic motion applied in each of three orthogonal planes; the frequency of the harmonic motion shall be varied uniformly from 5 to 500 Hz and return to 5 Hz in one minute; the amplitude shall be 0.2-inch double-amplitude; this frequency sweep shall be repeated 10 times in each plane. Failure to meet requirements of 4.6 or 3.5.1 shall constitute failure of this test.

4.6.2 Temperature-altitude. The eyepiece shall be tested in accordance with MIL-STD-810, Method 504.1, with modified temperature and altitude conditions as specified herein. These conditions are: Operating temperature +52°C and -32°C, nonoperating temperature +71°C and -35°C. Operating altitude 15,000 feet, storage altitude 50,000 feet. The following test sequence shall be followed:

TABLE VI. Test sequence.

MIL-STD-810 Equipment category	Procedure step	Altitude (feet)	Temperature (°C)	Inspection Point
3	1-b	Site	-32	No
3	3	15,000	-32	Note 1
5	3	50,000	-35	Note 1
3	5	Site	Site	No

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TABLE VI. Test sequence. - (Continued)

MIL-STD-810 Equipment category	Procedure step	Altitude (feet)	Temperature (°C)	Inspection Point
3	6	Site	+52	No
3	10	15,000	+52	Note 1
6	10	50,000	+71	Note 1
6	15	Sit	Site	Check for damage

NOTE 1 Eyepiece shall be inspected for internal moisture, fogging or frosting at the completion of the step. Failure to satisfactorily perform the listed inspections during test of 4.6 or Table VI, shall constitute failure of this test.

4.6.3 Temperature shock. The eyepiece shall be tested in accordance with MIL-STD-810, Method 503.1 with the temperature conditions of 3.5.3. The eyepiece shall be held at the temperature extremes for 2 hours in each step instead of 4 hours. Step 9 shall be limited to a visual check for damage.

Failure to meet requirements of 4.6 or 3.5.3 shall constitute failure of this test.

4.6.4 Humidity. The eyepiece shall be tested in accordance with MIL-STD-810, Method 507.1, Procedure I. The input to the eyepiece may be protected with coating on the lens only or be tested installed into a monocular. Step 1 shall consist of pre-inspection for damage (see 3.9.1). Step 6 shall consist of post inspection for damage. Step 7 shall consist of post testing required in paragraph 4.6. Failure to meet the requirements of 4.6 or 3.5.4 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The eyepiece is used with the AN/AVS-6(V)1, AN/AVS-6(V)2, in military helicopters as a pilot's aid to vision during nighttime, low-level, and nap-of-the-earth flight.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Time frame required for submission of IPT number of assemblies required and disposition of IPT samples (see 4.3).
c. Necessary action by contractor in event of Group C failures.
d. That data item DI-T-1903, DI-T-1906 and supplements thereto shall be a part of any contract incorporating this specification to guard against the use of material with hazardous levels of radioactivity.
e. Group B and Group C sampling plans. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
f. MIL-STD-810C shall be used for environmental tests of section 4.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

6.3 Definitions. See 3.9.

6.4 Subject term keyword listing.

ANVIS
Eyepiece
Lens Assembly

Custodian
Army - CR

Preparing Activity
Army - CR

Project 5855-A304

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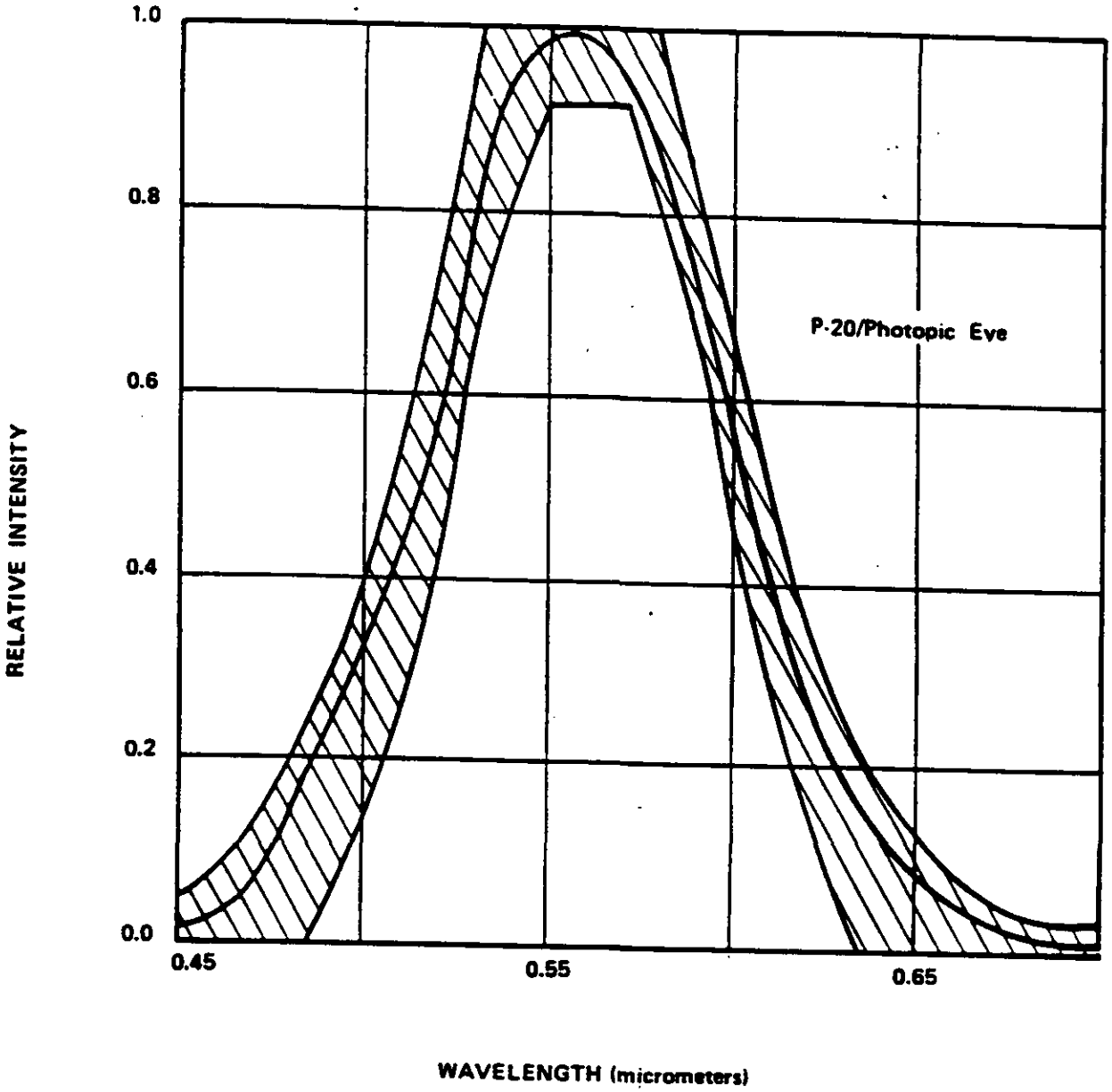


FIGURE 1. Spectral distribution of P-20/photopic eye combination.

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MTF SPECIFICATION	MODULATION, PERCENT REQUIREMENTS							
	RADIAL				TANGENTIAL			
	10LP/MM	20LP/MM	30LP/MM	40LP/MM	10LP/MM	20LP/MM	30LP/MM	40LP/MM
ON AXIS CENTERED PUPIL	92	83	75	70	92	83	75	70
ON AXIS VERTICAL DECENTERED PUPIL +2MM DECENTER	90	80	73	65	90	80	73	65
ON AXIS VERTICAL DECENTERED PUPIL +2MM DECENTER	86	70	52	37	88	75	64	57
OFF AXIS (2/3 FIELD) CENTERED PUPIL	90	78	69	64	90	78	60	48
OFF AXIS (2/3 FIELD) VERTICAL DECENTERED PUPIL +2MM	89	76	65	56	89	70	56	40
OFF AXIS (2/3 FIELD) VERTICAL DECENTERED PUPIL +5MM	85	55	35	18	80	45	25	5
OFF AXIS (2/3 FIELD) HORIZONTAL DECENTERED PUPIL +2MM	92	82	75	70	90	75	50	35
OFF AXIS (2/3 FIELD) HORIZONTAL DECENTERED PUPIL -2MM	90	80	73	70	90	76	66	55
OFF AXIS (2/3 FIELD) HORIZONTAL DECENTERED PUPIL +5MM	90	80	72	64	88	72	60	48
OFF AXIS (2/3 FIELD) HORIZONTAL DECENTERED PUPIL -5MM	90	74	64	55	80	60	40	25

FIGURE 2. MTF REQUIREMENTS

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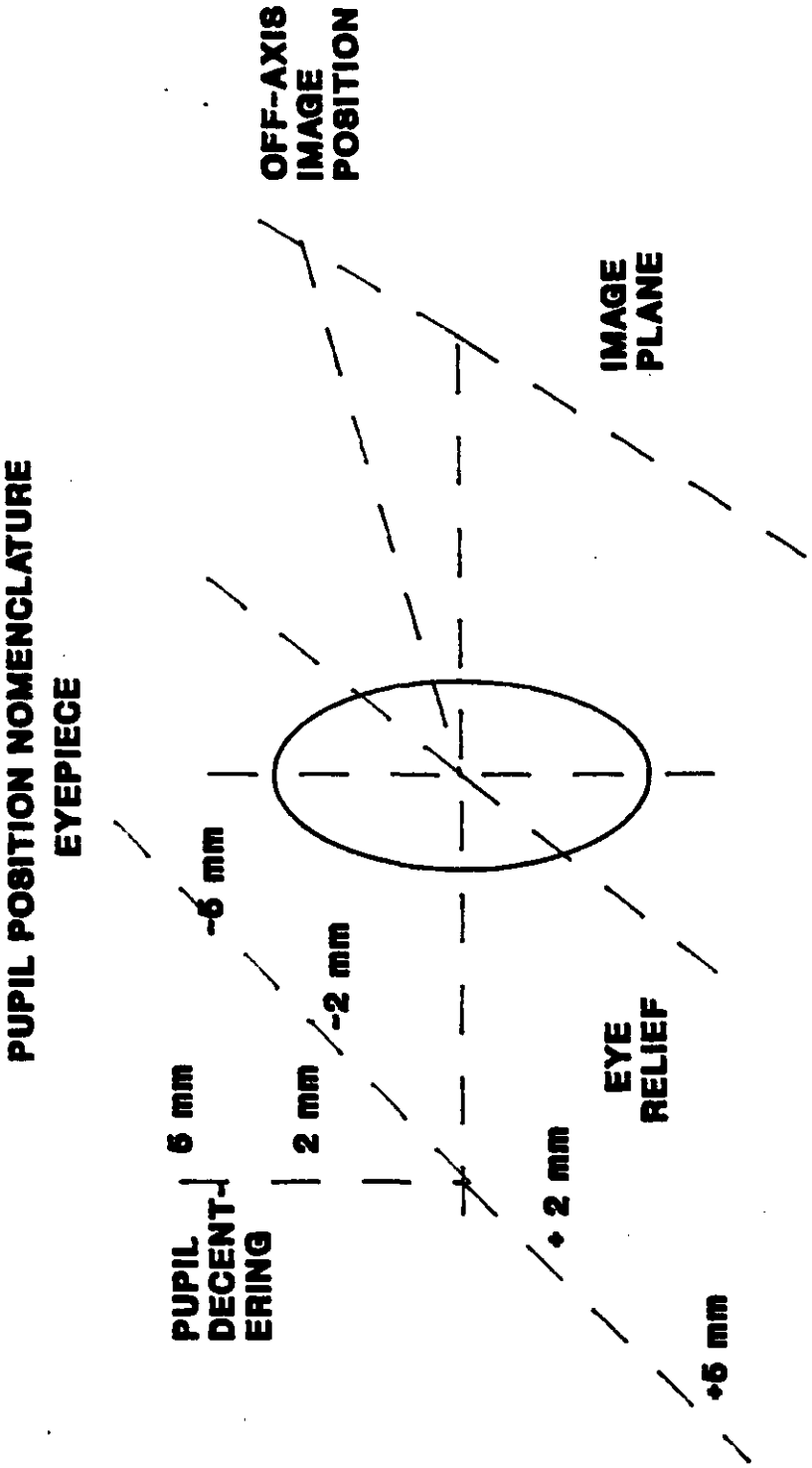


FIGURE 2. MTF REQUIREMENTS-Continued

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PINHOLE		SLIT
lp/mm	MTF,%	MTF,%
0.000	100.000	100.000
2.000	99.500	99.589
4.000	98.000	98.363
6.000	95.600	96.340
8.000	93.000	93.549
10.000	89.000	90.032
12.000	84.000	85.839
14.000	78.500	81.033
16.000	73.000	75.683
18.000	67.000	69.865
20.000	60.000	63.662
22.000	54.000	57.162
24.000	46.900	50.455
26.000	40.000	43.633
28.000	33.000	36.788
30.000	26.000	30.011
32.000	19.000	23.387
34.000	13.500	17.001
36.000	8.000	10.929
38.000	4.000	5.242
40.000	0.000	0.000

PINHOLE DIAMETER - 0.0305
SLIT WIDTH - 0.0250

FIGURE 3 MODULATION TRANSFER FUNCTION

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